

Final Projects

Mobile Systems

MobSys 2026

Deadline: not later than Sun. evening 27.09.2026

Projects can be completed and submitted earlier.

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1. Project Overview: Topic and Group Size

- Choose your own project idea
- Define an appropriate project scope
- Maximum of 4 participants per group
 - The larger the group, the larger and more complex the project should be
- Actuators, sensors, user interfaces, etc. may extend beyond the scope of the exercise (e.g., Bluetooth or NFC communication)

2. Development Environment

Framework

- Android - App
 - Android 14 .. 16 (API 34 .. 36)
 - (API **34**) as in the exercises
- Running applications in the simulator and optionally on a mobile phone
- Development environment:
 - Android Studio 2025 / 2026
 - **Kotlin Source Code**

3. Content

Activities and sensors

- At least 3 Activities:
 - At least one transfer of **data to next Activity and back;**
(Data exchange between the Activities, Intents)
 - For group of 3: at least +1 Activity
 - Pay a little attention to the user experience think about the development process
 - Include a CustomView inside an Activity
 - Use a RecyclerView in at least one of these Activities
- Sensors
 - Reading and using two sensors
(If necessary in a separate activity)

3.5 Possible content

- UI, Layouts, Events & Listeners
 - Diversify the usage of UI Components. Use widgets besides Buttons and TextViews. Like ProgressBar, RatingBar, ToolBar, CheckBox, Switch and so on
- CustomViews
- ListView / RecyclerView, Custom List Adapters
- Activities, Fragments
- Intents (Explicit & Implicit)
 - Camera, Geo View, Pickers (Contacts, Gallery, ...)
- Animations, Canvas Drawing, Charting, Graphing, Games
- Sensors: Accelerometer, GPS, Camera, ...
- Data Storage:
 - Shared Preferences, Persistence Frameworks (e. g. Google Room), Databases, SQLite
- Data Exchange
 - Networking: Wifi, Bluetooth or NFC
 - HTTP: Public REST-API Usage (e. g. IMDB, Weather)

4. Delivery and timeline

Submission of the following content is required **by 27 September 2026**

1. PDF with short description of the project Screenshots of the application (approx. 1-4 pages)

1.1 Title of the App

1.2 Who are the participants in the project, names + matrikel

1.3 What does the App do

1.4 How does the App look like (screenshots with short descriptions)

1.5 Brief comments on problems or special notes, if applicable

2. Project files packed together as rar archive

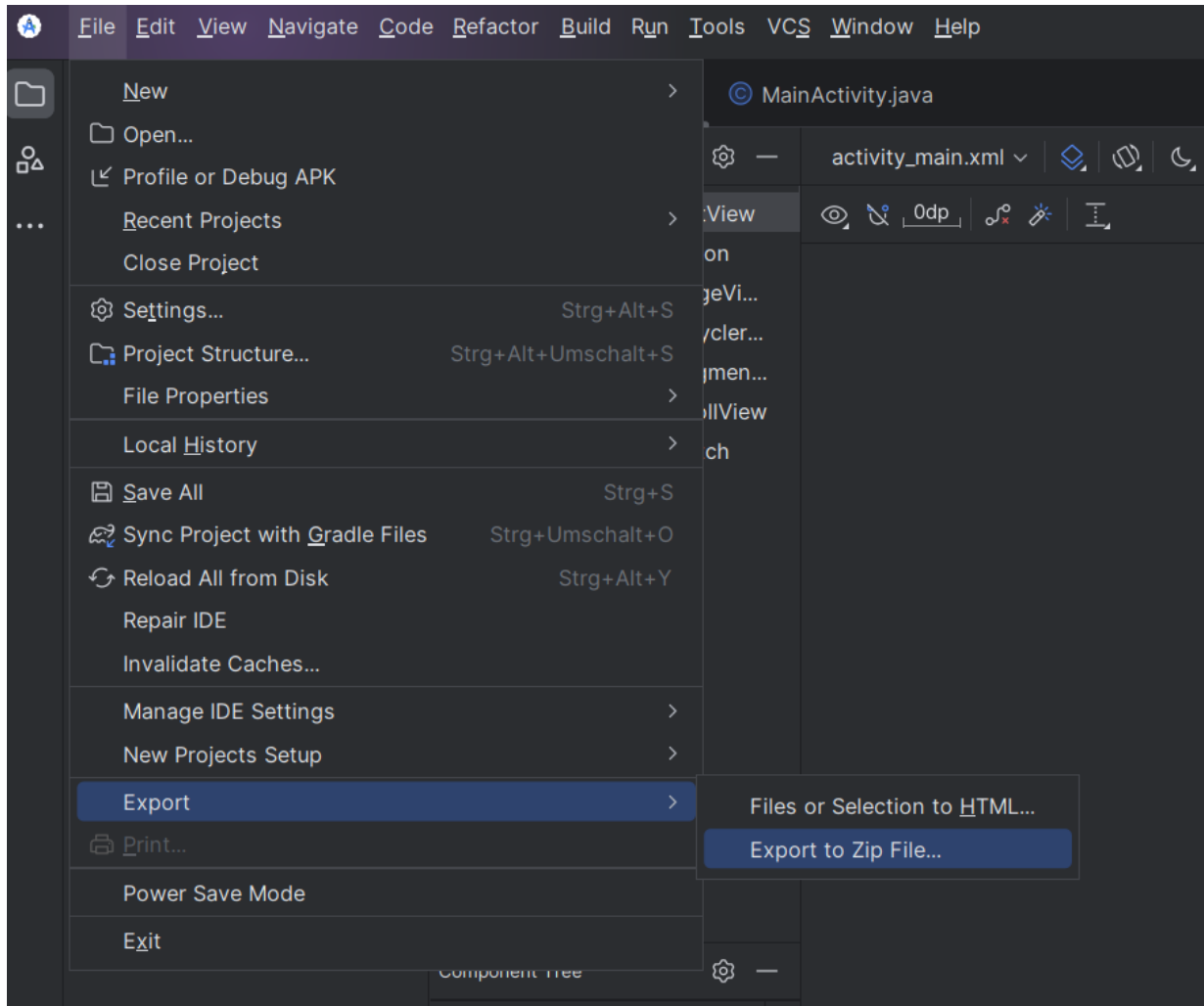
2.1 If applicable, also on **Google Drive** or GitHub / GitLab

3. Sending by email (google drive link)

4. Additionally an email without a file / link, so that I can consider intercepted emails

Easy way to send your project

Export your project to a Zip-File



Build a new archive / RaR-archive
Or upload the Zip-File to google drive

4.1 Notes on the project files

Copying files manually

- The project folder should be found, similar to mine, in the folder under: **"c:\Users\admin\AndroidStudioProjects\"**
- The project folder contains all sources and temporary data
- e.g.:

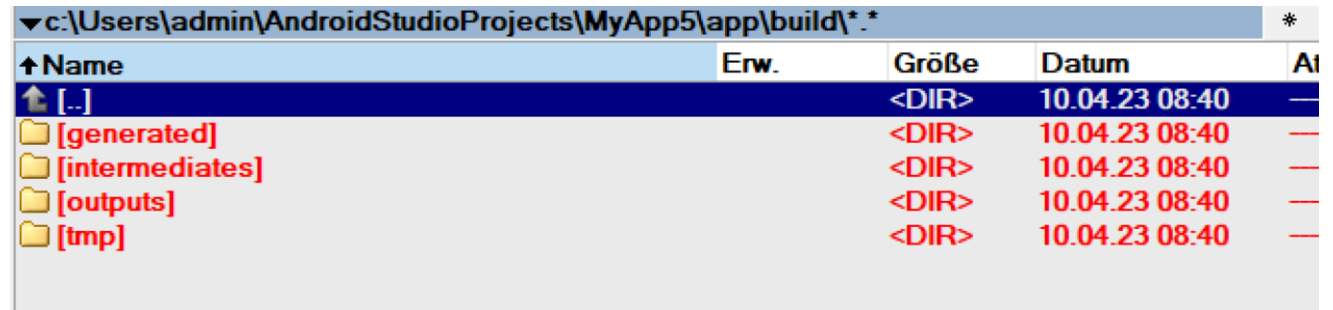


The screenshot shows a Windows File Explorer window with the address bar set to `c:\Users\admin\AndroidStudioProjects\MyApp5\*`. The window displays a list of files and folders. The table below represents the data shown in the file list.

↑ Name	Erw.	Größe	Datum
↑ [.]		<DIR>	10.04.23 08:37
[.gradle]		<DIR>	10.04.23 10:21
[.idea]		<DIR>	10.04.23 10:20
[app]		<DIR>	10.04.23 08:40
[gradle]		<DIR>	10.04.23 08:37
.gitignore		225	10.04.23 08:37
build	gradle	229	10.04.23 08:37
gradle	properties	1.267	10.04.23 08:37
gradlew		5.766	10.04.23 08:37
gradlew	bat	2.763	10.04.23 08:37
local	properties	449	10.04.23 08:37
settings	gradle	323	10.04.23 08:37

4.2 Project files

- The temporary files and folders in the project path
"...\\app\\build" can be deleted, these contain only intermediate results for the translations (approx. 40 MB)



↑ Name	Erw.	Größe	Datum	At
↑ [.]		<DIR>	10.04.23 08:40	—
📁 [generated]		<DIR>	10.04.23 08:40	—
📁 [intermediates]		<DIR>	10.04.23 08:40	—
📁 [outputs]		<DIR>	10.04.23 08:40	—
📁 [tmp]		<DIR>	10.04.23 08:40	—

- When deleting, do not forget to make a backup copy beforehand, in case errors occur during deletion